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TriCvT-DTI: Predicting Drug–Target Interactions Using Trimodal Representations and Convolutional Vision Transformers

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***Abstract*— Predicting interactions between drugs and their targets is vital for drug discovery and repositioning. Conventional techniques are slow and labor-intensive, while deep learning algorithms offer efficient solutions. However, deep learning often focus on single drug representations or simplistic combinations, leading to suboptimal feature representation. Moreover, the prevalent use of convolutional neural networks (CNNs) in drug image representation neglects the necessity for both local and global drug information in Drug-Target Interaction (DTI) tasks. To address these challenges, we propose TriCvT-DTI, a novel approach that combines molecular images, chemical sequence features, and graph representations of drugs to comprehensively capture structural, spatial, and functional aspects. TriCvT-DTI introduces a bidirectional multi-head attention mechanism for interactive feature learning between drugs and targets, enhancing performance by modeling complex relationships. By using Convolutional Vision Transformers (CvTs), TriCvT-DTI can effectively extract structural and spatial features from drug images. We evaluate our model on three datasets: Human, C. elegans, and Davis, and we compare it with state-of-the-art methods. Then we train TriCvT-DTI with uni-modality and bi-modality to compare then extract the impact of each modality on TriCvT-DTI. Experimental results demonstrate that TriCvT-DTI outperforms existing methods on both balanced and unbalanced datasets. Moreover, it presents impressive generalization capabilities on the Drug-Target Interaction (DTI) task.**

***Index Terms*— Bioinformatics, Computational drug discovery, Drug-Target Interactions, Medications, Convolutional Vision Transformer, Deep Learning.**

I. Introduction

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DRUG discovery is a complex, costly, and time-consuming process, with an estimated average cost of 2.6 billion dollars and over 10 years from initial research to market approval [1]. Infectious diseases like COVID-19 have underscored the importance of computational drug discovery, where AI models expedite the identification of drug candidates, optimize treatment strategies, and enhance understanding of disease mechanisms. These advancements address the challenges of limited knowledge about disease mechanisms and ethical issues in drug trials [2].

Drug-target interactions (DTIs) are crucial in drug discovery, as drugs need to interact with disease targets to exert their therapeutic effects. Conventional techniques for predicting DTIs [3] involve biochemical tests like affinity chromatography and radioligand binding assays, along with high-throughput screening (HTS). These techniques are useful for detecting potential interactions, but they can require a lot of resources and may not comprehensively address all possible drug-target pairs. Advances in computational drug discovery are helping to overcome these limitations by providing more efficient and comprehensive methods for predicting DTIs, ultimately improving the efficiency and effectiveness of the drug discovery process.

Machine learning (ML) has gained traction in DTI prediction, leveraging known interactions to develop models that predict new DTIs [4]. ML algorithms extract patterns from diverse data sources, such as molecular structures and interaction networks, making drug screening faster and more cost-effective. Techniques like support vector machines (SVMs) [5], random forests [6], and deep learning methods have shown promise. Deep learning models, including CNNs [7], GNNs [8], and attention-based mechanisms [9], have demonstrated improved accuracy. However, traditional ML often struggles with feature representation, necessitating advances in encoding molecular and protein data for better performance.

While most deep learning models use uni-modality (SMILES sequences, molecular graphs, or molecular images) or bi-modality (two drug representations), these approaches may not fully capture drug characteristics and interactions. Tri-modality representations offer a more holistic understanding, potentially enhancing performance and robustness across datasets.

Existing methods often focus on extracting features from drugs and targets separately, leading to suboptimal integration of interaction information. By incorporating convolutional neural networks (CNNs) and Vision Transformers (ViTs) for molecular image representation, we aim to extract local features, long-range dependencies, and 3D interaction patterns. Convolutional Vision Transformer (CvT) addresses these needs by capturing critical high-level, non-local, and 3D features, advancing drug discovery research.

In this paper, we propose TriCvT-DTI, a novel drug-target interaction (DTI) prediction model that effectively integrates tri-modality representations: molecular images, chemical sequences, and graph drug representations. Unlike existing approaches that rely on uni-modal or bi-modal representations, our tri-modal framework captures a more comprehensive and nuanced understanding of drug features, leading to improved robustness and performance across diverse datasets. To extract features from molecular images, we leverage the Convolutional Vision Transformer (CvT), which excels in capturing not only local and global features but also high-level, non-local, and 3D interaction patterns critical for accurate DTI modeling. This allows our model to go beyond traditional convolutional neural networks (CNNs), providing enhanced feature representation capabilities.

TriCvT-DTI's key contributions lie in: (1) Its innovative use of tri-modal representations (molecular images, chemical sequences, and graph) to comprehensively capture structural, spatial, and functional information critical for drug–target interaction (DTI). This holistic integration significantly enhances feature representation and demonstrates strong generalizability across datasets of varying sizes and complexities. (2) The incorporation of the Convolutional Vision Transformer (CvT), which not only captures local features but also excels in extracting high-level, non-local, and 3D interaction patterns critical for accurate DTI modeling. This addresses longstanding challenges in the field by improving prediction performance and capturing complex molecular relationships. (3) Highlighting the impact of different modal information. By comparing the AUC value (Area Under Curve) with every modality weight, we can identify the complementary strengths of each modality, demonstrating the significant contribution of tri-modality.

Furthermore, this approach demonstrates remarkable adaptability and robustness across diverse datasets and tasks, showcasing its potential to accelerate drug screening processes and elevate success rates in drug discovery endeavors.

The remainder of this article is structured as follows: Section II introduces the research progress in uni-modality and bi-modality methods for drug-target interaction prediction with the limitation of every research. Then, Section III describes TriCvT-DTI approach and the reason to use every part. However, in Section IV, outlines the experimental setup, including datasets, metrics, experiment settings and also our experiments with the results obtained. Next, the analysis and discussion, focusing on the performance of TriCvT-DTI, and the impact of different modal information, are presented in Section V. Finally, we conclude our paper in Section VI.

II. Literature Review

Over the past decades, computational methods for drug-target interaction (DTI) prediction have evolved, including docking-based, ligand-based, and chemogenomics-based techniques [10]–[14]. Docking methods require 3D structures of proteins, limiting applicability when unavailable. Ligand-based methods depend on known ligands, which may be sparse. Chemogenomics approaches utilize drug and target data from databases, offering broader versatility.

A. Convolutional Vision Transformer (CvT)

The Convolutional Vision Transformer (CvT), introduced in 2021 [15], combines CNNs and Vision Transformers to process multi-resolution features for visual tasks. Its ability to capture both local and global features makes it effective for molecular image analysis, aiding in accurate DTI prediction. CvT's framework is detailed in Section III-A.

B. Related Methods

Machine learning and deep learning have become central to DTI prediction [13], [16]–[20]. Lan et al. [21] introduced PUDT using positive-unlabeled learning with methods like Random Walk and SVM for classification. Other approaches include network-based [22]–[24], clustering-based [25], kernel-based [26], and matrix factorization [27]–[29]. Deep learning techniques often focus on encoding drugs and targets for binary classification tasks, utilizing sequence, graph, and image methods.

C. Uni-modality Methods

1) Sequence-based Methods: Many DTI prediction models focus on sequences. Öztürk et al. [30] introduced DeepDTA, which uses CNNs to extract features from SMILES and protein sequences. Other models like MATT-DTI [31] and ELECTRA-DTA [32] enhance feature extraction using self-attention mechanisms. Despite their success, sequence-based models often neglect molecular structures and atomic interactions. Ingoo Lee et al. [33] addressed this by proposing DeepConv-DTI, which captures local features using CNNs and drug fingerprints.

2) Graph-based Methods: Graph neural networks (GNNs) are increasingly used to model molecular graphs for DTI prediction. Graph-CPI [34] represents drugs as graphs and uses GNNs for learning features. Other models, like GraphDTA [8] and DGraphDTA [35], improve DTI prediction by modeling atomic interactions. GNN-CPI [36] fuses graph features to predict compound-protein interactions. However, graph models often lack details like double and triple bonds, suggesting the need for multi-modal approaches.

3) Image-based Methods: Image-based methods are less common in DTI prediction but have shown promise in enhancing feature extraction. PWO-CPI [7] and DTI-CNN [37] use CNNs to process molecular images, improving interaction prediction. CAT-CPI [38] and DEEPScreen [39] integrate CNNs with transformers and CNNs, respectively, to process drug images. However, image methods struggle with interpretability and may lack essential chemical details.

D. Bi-modality Methods

Many methods focus on a single modality, such as SMILES or molecular graphs. Incorporating multiple modalities improves representation accuracy, as seen in MCL-DTI [40], which combines chemical text and 2D images. TriCvT-DTI integrates three modalities (images, graphs, and chemical sequences) providing comprehensive molecular representation and improved DTI predictions. The model uses CvT for enhanced feature extraction from molecular images, outperforming traditional CNN-based methods.

III. TriCvT-DTI

In drug-target interaction (DTI) prediction, TriCvT-DTI frames the task as binary classification, predicting whether a drug and target will interact (1) or not (0). TriCvT-DTI utilizes tri-modality as inputs (image, chemical sequence, and graph representations) of the drug, and the target FASTA sequence. The architecture consists of four modules: the feature encoder, feature decoder, feature fusion, and classifier. The input data is processed using RDKit [41] to extract drug images, chemical sequence features, and graph representations from SMILES. TriCvT-DTI represents drugs as molecular graphs, capturing atomic bonds and spatial arrangements, while simultaneously encoding chemical sequences to capture sequential patterns. These modalities are integrated to enhance DTI prediction.

A. Convolutional Vision Transformer (CvT): Mathematical explanation

The Convolutional Vision Transformer (CvT) combines convolutional operations with Transformer mechanisms. CvT integrates Convolutional Token Embedding (CTE), Convolutional Projection, Multi-Head Self-Attention (MSA), Layer Normalization (LN), MLP Block, and Residual Connections.

CTE transforms the input image via convolutional operations (Eq. 1):

$$CTE(x_{i-1}) = \mathrm{Conv2D}(x_{i-1}; k, s) \tag{1}$$

where x_{i-1} is the input from the previous layer, Conv2D is the convolution operation, k is the kernel size, and s is the stride.

The Convolutional Projection further refines the representation (Eq. 2):

$$x^v = \mathrm{Flatten}(\mathrm{Conv2D}(\mathrm{Reshape2D}(x_i), s)) \tag{2}$$

where x^v is the token input for the query, key, and value matrices, and Reshape2D reshapes the input to a 2D map.

MSA enhances feature relationships, followed by residual connections and LN (Eq. 3):

$$\mathrm{LN}(x) = \frac{(\mathrm{MSA}(x^v) + x_i - \mu)}{\sigma} \cdot \gamma + \beta \tag{3}$$

where μ and σ are the mean and standard deviation of the input, and γ and β are learnable parameters.

The final output combines the MSA and MLP blocks (Eq. 4):

$$\mathrm{Output} = \mathrm{MLP}(\mathrm{LN}(x)) + \mathrm{MSA}(x) \tag{4}$$

These steps allow CvT to extract both local and global features, improving predictive capabilities for drug discovery.

(A) **The Feature Encoder Module:** TriCvT-DTI extracts drug features by integrating image, chemical sequence, graph features of drugs, and the FASTA sequence for targets.

1) Image Modality: To capture both local and global features of drug images and enhance drug-target interaction prediction, TriCvT-DTI employs a Convolutional Vision Transformer (CvT) backbone, detailed in Section III-A, combining convolutional operations and Transformer mechanisms as introduced in Section II-A. This integration strengthens predictive performance in drug discovery. Drug structural formula images are generated from SMILES sequences using RDKit software [41], offering visual molecular representations. These images encode structural information like bond connectivity, aromaticity, and stereochemistry, which are challenging to derive from text. Visualizing these features provides a more intuitive molecular understanding.

The input image is represented as $\mathcal{I} \in \mathbb{R}^{n\times n}$, where n denotes the image size. Processing through the CvT backbone produces the image feature map of the drug, as shown in Eq. 5:

$$x_{\text{img}} = \mathcal{CVT}(\mathcal{I}) \tag{5}$$

2) Chemical Sequence Modality: Chemical sequence features provide additional information about drug molecules, consisting of two components: feature type and feature family [41]. These classify characteristics like hydrogen bond donors and aromaticity, aiding pharmacophore matching. Using SMILES sequences, we extract chemical text features with RDKit [41], selecting relevant feature family/type data and associated atoms, as shown in Fig. 2.

To process the drug text, we apply the k-gram method, segmenting sequences into phrases of length k. This forms a dictionary of phrases, replacing words with numerical values based on dictionary positions. These values are embedded into the model, producing x_{text}, the chemical sequence feature output. Fig. 3 illustrates the k-gram method with $k = 3$, showing segmentation and embedding. This approach efficiently encodes textual drug information, improving representation and analysis.

3) Graph Modality: To leverage the power of graph representation in DTI, we preprocess drugs by converting their SMILES representations into graph structures using RDKit [41]. Each drug is represented as a graph $G = (V, E)$, where nodes V represent atoms and edges E represent bonds. Node features include atomic number, chirality, hybridization, degree, formal charge, aromaticity, and ring membership, while bond features include type, stereochemistry, and conjugation. These features enable the Graph Convolutional Network (GCN) to differentiate atom types and their roles in the graph.

We use a GCN model proposed by Kipf and Welling [42], which operates on graph structures using a node

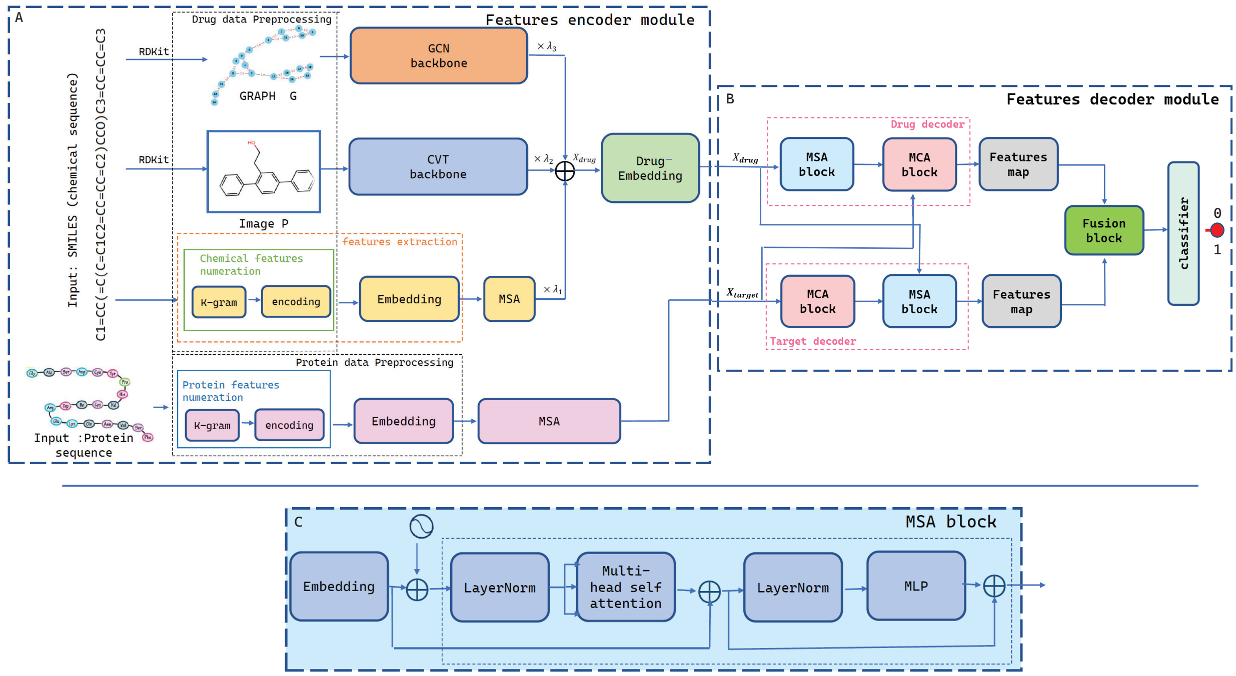

Fig. 1. The architecture of the proposed TriCvT-DTI approach. (A) The architecture of TriCvT-DTI features encoder. (B) The architecture of TriCvT-DTI features decoder. (C) The architecture of the MSA Block.

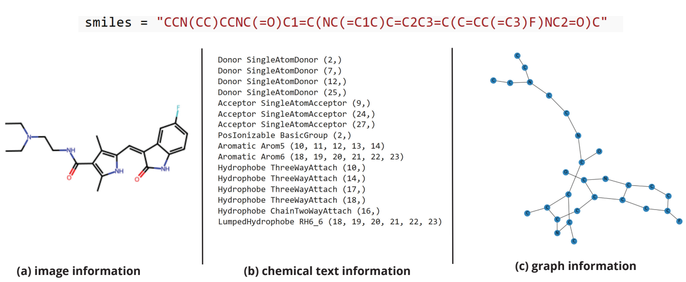

Fig. 2. Tri-modal representation of 5-fluorouracil (5-FU), used in cancer treatment. (a) Drug image. (b) Chemical text. (c) Drug graph.

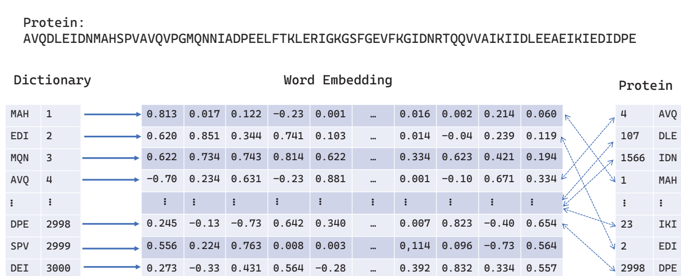

Fig. 3. Encoding protein sequence using the 3-gram method (segmentation and embedding).

feature matrix and an adjacency matrix. The GCN has three graph convolution layers followed by a global max pooling layer. During each graph convolution, node features are updated by applying a shared weight matrix and aggregating information from neighboring nodes via the adjacency matrix. This message-passing mechanism captures both local connectivity and atomic properties, producing enriched node embeddings at each layer.

To obtain a graph-level representation, the global max pooling layer aggregates the learned node-level features into a single representation vector, summarizing distant yet critical relationships within the molecule. By combining feature encoding, message passing, and global pooling, the GCN effectively handles the complexity of diverse node types and interactions in drug graphs. The final output, x_{graph}, represents the drug graph features:

$$x_{\text{graph}} = GCN(G) \tag{6}$$

We integrate multiple modalities of drug features to create a comprehensive representation of the compound, denoted as x_{drug}. These modalities include image features x_{img} (Eq. 5), chemical sequence features x_{text}, and graph-based features x_{graph} (Eq. 6). To emphasize the contribution of each modality, we combine these features using learnable weights (λ_1, λ_2, and λ_3). The drug feature vector x_{drug} is calculated as the weighted sum of these modalities:

$$x_{\text{drug}} = \lambda_1 \times x_{\text{text}} + \lambda_2 \times x_{\text{image}} + \lambda_3 \times x_{\text{graph}} \tag{7}$$

B. Target encoder

For the target sequence, the architecture is shown in Fig. 1.A, we start by using its FASTA sequence directly. We process this sequence using the k-gram technique to generate its embedding representation. Following this, a MSA module is employed to derive high-level features from the sequence, capturing complex dependencies and interactions within the sequence (the MSA block architecture is shown in Fig. 1.C). While the data is primarily sequence-based, the self-attention mechanism helps infer structural and functional relationships, enhancing the prediction of drug-target interactions by integrating both sequence and indirect structural insights.

The feature vector for the target sequence, denoted as x_{target}, is computed using this approach. Specifically, the embedding representation is obtained through the k-gram method, and the abstract features are extracted using the MSA module. This methodology allows for a comprehensive understanding of the target sequence, combining sequence-based and inferred structural information for improved prediction accuracy.

C. Feature Decoder Module

The feature decoder extracts drug-target interaction information through two decoders, each containing a Multi-Head Self-Attention (MSA) block and a Multi-Head Cross-Attention (MCA) block. Both blocks incorporate Layer Normalization (LN), MLP blocks, and residual connections, with the difference in attention computation. The MSA block captures internal feature dependencies, while the MCA block models interactions between drug and target features. In the MCA block of the drug decoder, drug and target features are input, and target features are linearly transformed to compute the attention output. This bidirectional learning process produces two feature maps: $Z_{\text{drug-target}}$ (from drug features in the target decoder) and $Z_{\text{target-drug}}$ (from target features in the drug decoder), as shown in Fig. 1.B.

D. Feature Fusion Block

The fusion block integrates the $Z_{\text{drug-target}}$ and $Z_{\text{target-drug}}$ feature maps by concatenating them along the channel dimension, followed by 2D and 1D convolutions, an MLP block, and a fully connected (FC) layer. The final prediction is given by:

$$CONV = \text{Conv1D}\left(\text{Conv2D}\left(\left[Z_{\text{drug-target}}, Z_{\text{target-drug}}\right]\right)\right)$$
$$Z = \text{FC}\left(\text{MLP}\left(CONV\right)\right)$$

E. Classifier

The classifier uses a cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N}\sum_{n=1}^{N}\left(y_n \log(P_n) + (1-y_n)\log\left(1-P_n\right)\right)$$

where N is the total number of samples and y_n denotes the true label. The Adam optimizer [43] is used to efficiently update the model parameters.

IV. Experiments

A. Datasets

We utilized three publicly available DTI datasets: Human [44], C. elegans [44], and Davis [45]. Each dataset consists of three primary components: (1) drug sequences represented as textual data, (2) target protein sequences also in textual form, and (3) a binary interaction label indicating whether a drug-target interaction exists. Table I provides detailed statistics regarding the drugs and targets included in each dataset. Both Human and C. elegans datasets are balanced with positive and negative samples, sourced from high-confidence biochemical databases such as DrugBank [46] and Matador [47]. For Davis, a total of 64 different drugs and 379 targets were considered, with positive DTI pairs defined by kd values $<$ 30 units. Dataset splitting followed an 8:1:1 ratio for train, validation, and test sets in Human and C. elegans datasets, while the division for Davis dataset adhered to the approach outlined in MolTrans [9]. Additionally, we utilized the Biosnap dataset [48] for Drug-Drug Interaction (DDI) prediction, containing 9,648 drugs and 81,194 samples, with approximately 50.5% of samples being positive instances.

TABLE I
THE OCCURRENCES OF DTI DATASETS

Datasets	Drugs	Targets	Samples	Positive Samples	Balanced
C. elegans	1716	1856	7509	3892	TRUE
Davis	64	379	10439	1428	FALSE
Human	2496	1919	6184	3364	TRUE

B. Metrics

We evaluated our models using ROC-AUC (Receiver Operating Characteristic-Area Under the Curve), PR-AUC (PrecisionRecall-Area Under the Curve). ROC-AUC was the primary metric, assessing performance across both positive and negative examples. PR-AUC focused on positive instances, particularly for imbalanced datasets, highlighting model robustness in identifying positives. Recall measured the percentage of correctly identified positive samples, indicating the model's ability to learn positive instances. All results are presented as mean values with standard deviations, offering an overview of model performance and variability.

C. Experiment settings

The experiments were conducted using PyTorch [49] with 100 epochs. The hardware included an Intel i9 10700f processor, 32GB RAM, and an RTX 4090 GPU. The initial learning rate was set to 1e-3, explored within [1e-3, 1e-4, 1e-5], with a decay rate of 0.8, tested within [0.5–0.9]. The batch size was 32, explored within [16, 32, 64], and the dropout rate was 0.1, evaluated within [0.1–0.5]. Hyperparameter optimization employed a grid search technique.

D. Tri-modality experiments

For the tri-modality experiments, we performed a comprehensive assessment of our TriCvT-DTI model by benchmarking it against a variety of existing approaches. We chose prominent, state-of-the-art methods, including MCL-DTI [40], CAT-CPI [38], MolTrans [9], Mutual-DTI [50], and HiGraphDTI [51], given their recognition and frequent application in the domain. To evaluate graph-based methods, we incorporated GNN-CPI [36] and HiGraphDTI [51]. In contrast, TransformerCPI [52], DeepDTA [30], and DeepConv-DTI [33] were used to assess the effectiveness of sequence-based techniques. Additionally, PWO-CPI [7] was included to evaluate our model's ability to utilize image-based data. Furthermore, MCL-DTI [40] was employed to compare TriCvT-DTI with bi-modality methods. This diverse selection allowed for a thorough evaluation of TriCvT-DTI across multiple modalities, providing valuable comparisons with established models and insights into its effectiveness for drug-target interaction prediction. The state of the art and our experimental results are summarized in Table II.

V. Analysis and Discussion

A. TriCvT-DTI Performance Across Datasets

Table II presents the results of TriCvT-DTI compared to the latest methods. The results demonstrate that TriCvT-DTI outperforms the current state-of-the-art methods.

1) **Human dataset**: TriCvT-DTI achieves a ROC-AUC of 0.989, outperforming models like MCL-DTI (0.987) and Mutual-DTI (0.984). Although HiGraphDTI has a slightly higher ROC-AUC of 0.993, TriCvT-DTI remains competitive. With a PR-AUC of 0.990, it surpasses models like GNN-CPI (0.973), Mutual-DTI (0.962), and MCL-DTI (0.989), and matches HiGraphDTI (0.993), demonstrating high precision. TriCvT-DTI also achieves a Recall of 0.962, matching the best values of DeepConv-DTI and PWO-CPI, showing its robustness in identifying true positives. Its balanced performance across all metrics makes it a strong alternative.

2) **C. elegans dataset**: TriCvT-DTI outperforms all other methods on this dataset, achieving the highest ROC-AUC of 0.993, surpassing MCL-DTI (0.992) and HiGraphDTI (0.985). It also leads in PR-AUC with 0.995 and has a Recall of 0.968, demonstrating its effectiveness in extracting multimodal data features and identifying true positives.

3) **Davis dataset**: TriCvT-DTI performs excellently with high ROC-AUC and Recall metrics, surpassing all latest methods. The PR-AUC, affected by the dataset's imbalance (1 positive to 19 negatives), remains competitive, highlighting the model's precision. TriCvT-DTI's high Recall further demonstrates its effectiveness in identifying true positives. Those results prove that TriCvT-DTI has a strong generalisation performance and it can suite well with all dataset even balanced or not balanced or big or small datasets because it can extract enough features to make the classification using three modalities.

B. Complementary Roles of Modalities in TriCvT-DTI

Text modality encodes chemical and biological properties, capturing information not found in graphs or images. Graph modality captures structural relationships within molecules and biological entities. Image modality encodes spatial information, such as molecular substructures and 3D conformations, revealing insights missed by text and graph data. By integrating all three modalities, TriCvT-DTI compensates for each other's limitations, improving prediction performance.

C. Impact of different modal information

In the TriCvT-DTI model, the parameters λ_1, λ_2, and λ_3 represent the relative contributions of the image, text, and graph modalities, respectively. These parameters are dynamically learned during training, allowing the model to adaptively adjust the contribution of each modality based on its relevance to the dataset and the task of predicting drug-target interactions.

Figures 4.A, 4.B, and 4.C show the relationship between AUC values at random epochs and the parameters λ_1, λ_2, and λ_3 across the C. elegans, Human, and Davis datasets, respectively, where the AUC is greater than or equal to 0.90. This analysis provides insights into the optimal and stable parameter values, enhancing our understanding of their impact on model performance. For the Human dataset (Fig. 4.B), the optimal values are $\lambda_1 = 0.61$, $\lambda_2 = 0.66$, and $\lambda_3 = 0.37$, with the text modality (λ_2) contributing most, followed by the image (λ_1), and the graph modality (λ_3) contributing least but still significantly.

For the C. elegans dataset (Fig. 4.A), the optimal values are $\lambda_1 = 0.57$, $\lambda_2 = 0.70$, and $\lambda_3 = 0.21$. Here, the text modality (λ_2) dominates, with the graph modality contributing less and the image modality moderately.

For the Davis dataset (Fig. 4.C), the optimal parameters are $\lambda_1 = 0.75$, $\lambda_2 = 0.68$, and $\lambda_3 = 0.67$, where the image modality (λ_1) contributes most, followed by text (λ_2) and graph (λ_3), indicating a more balanced contribution across all modalities.

These results show that while the text modality consistently has the highest weight, the relative importance of image and graph modalities varies by dataset. The Human dataset balances text and image features with graph modality as supplementary, while the C. elegans dataset heavily relies on text, with moderate image use and minimal graph contribution. The Davis dataset emphasizes image features while retaining significant contributions from both text and graph modalities. This adaptability highlights TriCvT-DTI's ability to optimize performance by adjusting modality weights according to dataset characteristics.

In summary, TriCvT-DTI's dynamic learning of modality contributions optimizes performance across datasets by adjusting the balance of image, text, and graph features.

VI. Conclusion

TriCvT-DTI represents a significant advancement in predicting drug-target interactions. By combining molecular images, chemical sequence features, and graph-based drug representations, it provides a more detailed understanding of these interactions, including structural, spatial, and functional aspects. The model has shown strong performance across various datasets, indicating its potential to greatly impact drug discovery and development.

There are several areas for improvement and exploration. Enhancing the model through optimization, alternative attention mechanisms, or additional data types could increase its accuracy and versatility. TriCvT-DTI could also be adapted to address specific challenges in drug discovery, such as drug repurposing or personalized medicine, offering new insights for different therapeutic areas. Ensuring that the model's predictions are interpretable will be essential for gaining trust in clinical settings. Additionally, investigating advanced techniques like self-supervised learning and multi-task learning

TABLE II
PERFORMANCE METRICS FOR DIFFERENT METHODS ON VARIOUS DATASETS.

Dataset	Approaches	ROC-AUC values	PR-AUC values	Recall values
Human	DeepDTA [30]	0.953 ± 0.002	0.981 ± 0.003	0.946 ± 0.024
	GNN-CPI [36]	0.974 ± 0.004	0.973 ± 0.005	0.953 ± 0.019
	DeepConv-DTI [33]	0.985 ± 0.001	0.982 ± 0.001	**0.962 ± 0.002**
	TransformerCPI [52]	0.971 ± 0.002	0.973 ± 0.002	0.942 ± 0.004
	PWO-CPI [7]	0.982 ± 0.003	0.980 ± 0.002	**0.962 ± 0.001**
	MolTrans [9]	0.978 ± 0.002	0.978 ± 0.001	0.933 ± 0.003
	MCL-DTI [40]	0.987 ± 0.001	0.989 ± 0.001	0.961 ± 0.002
	HiGraphDTI [51]	**0.993 ± 0.001**	**0.993 ± 0.001**	-
	Mutual-DTI [50]	0.984 ± 0.001	0.962 ± 0.019	0.943 ± 0.016
	TriCvT-DTI (Ours)	0.989 ± 0.002	0.990 ± 0.001	**0.962 ± 0.002**
C. elegans	DeepDTA [30]	0.987 ± 0.001	0.990 ± 0.002	0.964 ± 0.011
	GNN-CPI [36]	0.978 ± 0.002	0.975 ± 0.005	0.949 ± 0.003
	DeepConv-DTI [33]	0.980 ± 0.002	0.981 ± 0.001	0.937 ± 0.003
	TransformerCPI [52]	0.985 ± 0.001	0.985 ± 0.002	0.952 ± 0.002
	PWO-CPI [7]	0.979 ± 0.002	0.978 ± 0.003	0.933 ± 0.003
	MolTrans [9]	0.985 ± 0.001	0.984 ± 0.002	0.962 ± 0.001
	MCL-DTI [40]	0.992 ± 0.001	0.994 ± 0.001	0.959 ± 0.002
	HiGraphDTI [51]	0.985 ± 0.001	0.988 ± 0.001	-
	Mutual-DTI [50]	0.987 ± 0.004	0.948 ± 0.018	0.949 ± 0.013
	TriCvT-DTI (Ours)	**0.993 ± 0.001**	**0.995 ± 0.001**	**0.968 ± 0.002**
Davis	DeepDTA [30]	0.860 ± 0.002	0.238 ± 0.002	0.818 ± 0.003
	GNN-CPI [36]	0.840 ± 0.003	0.269 ± 0.003	0.696 ± 0.002
	DeepConv-DTI [33]	0.822 ± 0.002	0.192 ± 0.001	0.905 ± 0.003
	TransformerCPI [52]	0.841 ± 0.002	0.227 ± 0.001	0.842 ± 0.001
	PWO-CPI [7]	0.835 ± 0.002	0.188 ± 0.002	0.798 ± 0.002
	MolTrans [9]	0.907 ± 0.002	0.404 ± 0.002	0.800 ± 0.001
	MCL-DTI [40]	0.922 ± 0.001	0.492 ± 0.002	0.895 ± 0.002
	HiGraphDTI [51]	0.871 ± 0.003	0.547 ± 0.002	0.794 ± 0.003
	Mutual-DTI [50]	0.900 ± 0.002	**0.767 ± 0.002**	0.680 ± 0.001
	TriCvT-DTI (Ours)	**0.935 ± 0.001**	0.614 ± 0.001	**0.911 ± 0.001**

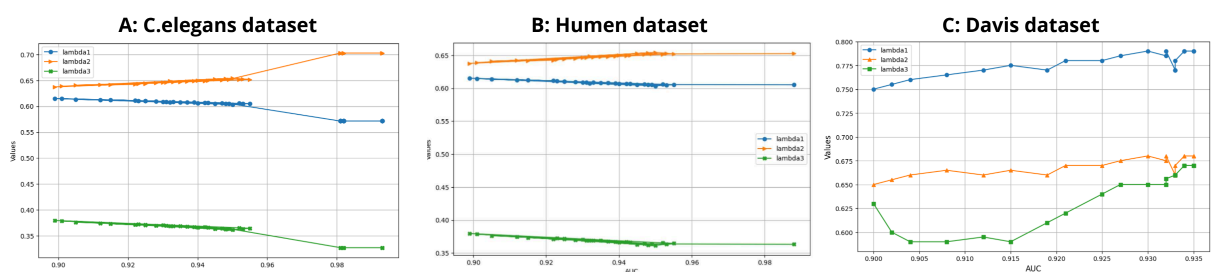

Fig. 4. Performance of AUC measure by the values of λ_1, λ_2 and λ_3. A: across C. elegans Dataset. B: across Human Dataset. C: across Davis Dataset.

may help better utilize large biological datasets and further improve TriCvT-DTI's effectiveness.

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